

Notes: Acemoglu, Akcigit, Alp, Bloom and Kerr (AER, 2018)

Innovation, Reallocation, and Growth

Contents

1 Big picture

2 Data and measurement (key objects)

2.1	Data sources
2.2	Sample construction
2.3	Why the sample is economically important
2.4	How R&D is measured for small firms
2.5	Size and age categories
2.6	Targeted moments
2.7	Examples from the targeted moment table

3 Models and empirical specifications (all equations + interpretation)

3.1	Preferences and final-good technology
3.2	Production and relative productivity
3.3	Firm heterogeneity
3.4	Innovation technology and R&D cost
3.5	Innovation step size
3.6	Entry and exit
3.7	Value functions and additive franchise values
3.8	Optimal innovation intensity and shutdown thresholds
3.9	Labor-market clearing
3.10	Aggregate growth
3.11	Welfare
3.12	Estimation by simulated method of moments

4	Identification and instruments (what makes the estimates credible)	
4.1	What identifies the structural parameters	
4.2	Why the selection mechanism is empirically central	
4.3	Validation through nontargeted moments	
4.4	Why reverse causality is not the main issue here	
4.5	What the paper can and cannot distinguish	
4.6	Why the policy message is nevertheless credible	
5	Tables and figures	
5.1	Table 1 and Table 2: calibrated and estimated parameters	
5.2	Table 3 and Figure 1: targeted moments and goodness of fit	
5.3	Table 4 and Figure 2: the baseline economy and the selection margin	
5.4	Table 5: nontargeted moments	
5.5	Figure 3 and Table 6: product-line distribution and growth decomposition	
5.6	Tables 7–9: simple industrial policies one at a time	
5.7	Tables 10–12: the social planner and optimal uniform policies	
5.8	Tables 13–19: robustness	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** How do innovation, firm selection, entry, exit, and the reallocation of skilled labor interact, and what kinds of industrial policies raise growth and welfare in such an environment?
- **Setting.** The paper builds a structural model of firm-level innovation with endogenous exit and heterogeneous innovative capacity, and then estimates it using U.S. Census microdata on output, employment, R&D, and patenting.
- **Core challenge.** Standard industrial policies often subsidize incumbent firms or their R&D. But in a world with heterogeneous firms, such policies may keep low-quality incumbents alive, absorb scarce skilled labor, and slow down growth instead of stimulating it.
- **Main conceptual contribution.** The paper puts *selection across firms* at the center of innovation policy. High-type firms and low-type firms use skilled labor differently, so policy matters not only for how much R&D is done, but *who* does it.
- **Key mechanism.** High-type firms are better innovators and therefore have a higher social value of staying active. Low-type firms are closer to the exit margin and use skilled labor less productively. Policies that free skilled labor from low-type operations and reallocate it toward high-type R&D generate growth.
- **Data contribution.** The estimation sample is built around continuously innovative manufacturing firms observed in 1987, 1992, and 1997, using five-year windows around each Census of Manufactures year. The sample contains 17,055 firm-year observations from 9,835 firms and covers more than 98% of industrial R&D in the United States during the sample period.
- **Main quantitative result.** In the baseline equilibrium, growth is 2.26% and welfare is normalized to 100. A social planner who can directly steer entry, exit thresholds, and innovation rates raises growth to 2.94% and welfare to 104.47.
- **Main policy result.** Uniform R&D subsidies to incumbents help only modestly. Subsidies to the operating costs of incumbents are actually harmful because they prop up low-type firms. The most powerful realistic uniform policy is a sizable tax on continued operation of incumbents.
- **Best uniform policy.** A tax of about 69% on the fixed costs of operation of incumbents raises growth to 2.54% and welfare to 101.42. The optimal joint uniform policy is similar: a large operation tax and a tiny tax on incumbent R&D.
- **Core lesson.** The distortion is not simply “too little R&D.” It is also *misallocated* R&D and too much skilled labor tied up in the operations of weak incumbent firms.
- **Economic takeaway.** Growth policy works best when it creates positive selection—moving resources away from low-innovation product lines and toward high-innovation firms—rather than when it indiscriminately subsidizes all incumbents.

2 Data and measurement (key objects)

2.1 Data sources

The paper combines four major U.S. microdata sources:

- the **Longitudinal Business Database (LBD)** for annual establishment-level employment,
- the **Census of Manufactures (CMF)** for detailed manufacturing output and operating variables every five years,
- the **Survey of Industrial Research and Development (RAD)** for firm-level R&D,
- and the **NBER Patent Database (PAT)** for patenting activity.

Interpretation. The paper needs all four because the model is about the joint evolution of *output*, *employment*, *R&D*, *patenting*, *entry*, and *exit*. No single dataset captures all those margins.

2.2 Sample construction

The operating years are 1987, 1992, and 1997, the CMF years. For each CMF year, innovation activity is measured in a five-year window around that year:

- 1985–1989 for the 1987 CMF,
- 1990–1994 for the 1992 CMF,
- 1995–1999 for the 1997 CMF.

A firm is called *innovative* if it either conducts R&D or files patents in the relevant window. The main sample is restricted to *continuously innovative* firms: if a firm is in operation in two CMF years, it must be innovative in both surrounding five-year windows.

- Total sample: 17,055 firm-year observations.
- Number of firms: 9,835.
- Coverage: more than 98% of industrial R&D over the period.

Interpretation. This restriction is important. The paper is not trying to model the decision to begin innovating from zero inside an ongoing firm. Instead, it wants a clean sample of firms whose growth, exit, and reallocation dynamics are tightly connected to innovation activity.

2.3 Why the sample is economically important

Although continuously innovative firms are only about 2% of manufacturing firms, they account for:

- about 50% of manufacturing employment,
- about 64% of manufacturing sales,
- about 58% of employment reallocation among continuing manufacturing firms,
- and about 65% of sales reallocation among continuing manufacturing firms.

The sample also captures most extreme growth episodes among small firms and entrants that later become very large.

Interpretation. The paper is not about the median tiny manufacturing plant. It is about the part of the economy where innovation and reallocation are quantitatively central.

2.4 How R&D is measured for small firms

RAD includes with certainty firms above a large R&D threshold, but smaller R&D firms are only sampled. To avoid missing innovative small firms, the paper uses patents to impute R&D for firms below the certainty threshold and not sampled in RAD.

Interpretation. The paper wants a model-consistent measure of innovative effort for the whole distribution, not only for large firms that are definitely surveyed.

2.5 Size and age categories

The targeted moments are built around coarse but economically meaningful bins:

- **Large** firms: more than 200 employees.
- **Small** firms: at or below that threshold.
- **Young** firms: 0–9 years old.
- **Old** firms: 10+ years old.

Firm age is measured from the first year the firm is observed in the LBD with positive employment.

2.6 Targeted moments

The SMM estimation targets 18 moments:

- exit rates by age-size bin,
- transitions from large to small and small to large,
- the probability that an entrant is small,
- employment growth by age-size bin,
- sales growth by age-size bin,
- R&D to sales by age-size bin,
- the five-year entrant share,
- the fixed-cost-to-R&D labor ratio,
- aggregate growth.

The aggregate growth moment gets five times the weight of the micro moments in estimation.

Interpretation. The moment choice mirrors the model logic. Exit moments pin down obsolescence and destruction. Growth moments pin down selection and innovation. R&D intensity moments pin down innovative capacity. Entry and transition moments pin down firm dynamics and the relative importance of entrants.

2.7 Examples from the targeted moment table

Table 3 reports the following model-versus-data comparisons:

- firm exit for small-young firms: 0.097 in the model versus 0.107 in the data,
- firm exit for large-old firms: 0.036 versus 0.036,
- employment growth for small-young firms: 0.101 versus 0.106,
- sales growth for small-old firms: 0.040 versus 0.024,
- R&D-to-sales for large-old firms: 0.059 versus 0.037,
- aggregate growth: 0.023 versus 0.022.

Interpretation. Even before seeing the policy experiments, the basic message is that the estimated model tracks the joint age-size-growth-exit patterns in the data surprisingly well.

3 Models and empirical specifications (all equations + interpretation)

3.1 Preferences and final-good technology

The representative household has CRRA utility

$$U_0 = \int_0^\infty e^{-\rho t} \frac{C(t)^{1-\vartheta} - 1}{1-\vartheta} dt. \quad (1)$$

Aggregate consumption is a CES aggregate over active product lines:

$$C(t) = \left(\int_{\mathcal{N}(t)} c_j(t)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}. \quad (2)$$

Interpretation. Consumers value variety. This matters because growth comes from innovation that improves the quality ladder and changes the set and quality of active product lines.

3.2 Production and relative productivity

A firm that operates product line j with leading-edge technology $q_{f,j}$ produces according to

$$y_{f,j} = q_{f,j} l_{f,j}, \quad (3)$$

but only after paying a fixed operating cost of ϕ units of skilled labor per active product line.

Relative productivity is defined as

$$\hat{q} \equiv \frac{q}{w_u}, \quad (4)$$

where w_u is the unskilled wage.

Interpretation. The fixed skilled-labor requirement is central. It means operating an existing product line absorbs scarce high-skill labor that could otherwise be used in R&D. That is the key scarcity margin in the policy analysis.

3.3 Firm heterogeneity

Firms come in two types:

$$\theta \in \{\theta_h, \theta_l\}, \quad \theta_h > \theta_l > 0. \quad (5)$$

High-type firms are better innovators. New entrants draw high type with probability α . High-type firms can transition to low type at flow rate ν .

Interpretation. This introduces *selection*. Young firms are more likely to be high type, but over time some high types decay into low types. The composition of active firms therefore changes endogenously with innovation, exit, and type transitions.

3.4 Innovation technology and R&D cost

If a firm of type θ_f with n_f product lines hires h_f skilled workers in R&D, its innovation flow is

$$X_f = \theta_f^\gamma n_f^\gamma h_f^{1-\gamma}. \quad (6)$$

Let $x \equiv X/n$ be innovation intensity per product line. Then the R&D cost function is

$$C(x, n, \theta) = w_s n x^{\frac{1}{1-\gamma}} \theta^{-\frac{\gamma}{1-\gamma}} = w_s n G(x, \theta), \quad (7)$$

where w_s is the skilled wage.

Interpretation. Two things matter for innovative output:

- the firm's type θ ,
- and how many product lines it already controls.

The first is pure innovative capacity; the second captures the Klette-Kortum style idea that larger firms have a broader base on which to innovate.

3.5 Innovation step size

A successful innovation improves the productivity of one product line by

$$q_j(t^+) = q_j + \lambda \bar{q}, \quad (8)$$

where $\bar{q}$ is average quality in the economy.

Interpretation. Each successful innovation is a proportional step up the quality ladder. The growth rate of the economy will therefore depend on both the step size λ and the frequency of creative destruction.

3.6 Entry and exit

Potential entrants choose innovation effort x_{entry} to solve

$$\max_{x_{entry} \geq 0} \{x_{entry} \mathbb{E} V_{entry}(\hat{q} + \lambda \bar{\hat{q}}, \theta) - w_s G(x_{entry}, \theta_E)\}. \quad (9)$$

Exit happens for three reasons:

- exogenous destruction at rate φ ,
- creative destruction by other innovators,
- endogenous obsolescence: the firm voluntarily shuts down low-productivity product lines.

Interpretation. The endogenous exit margin is what makes industrial policy nontrivial. A subsidy to keeping firms alive may preserve some product lines, but it may also preserve exactly the wrong ones.

3.7 Value functions and additive franchise values

Let $\tilde{V}_l(\hat{\mathbb{Q}})$ and $\tilde{V}_h(\hat{\mathbb{Q}})$ denote normalized firm values for low- and high-type firms. The paper shows that firm value is additive in franchise values:

$$\tilde{V}_k(\hat{\mathbb{Q}}) = \sum_{\hat{q} \in \hat{\mathbb{Q}}} \Upsilon_k(\hat{q}), \quad k \in \{l, h\}. \quad (10)$$

For each type there is an endogenous obsolescence threshold $\hat{q}_{k,\min}$ below which the product line is shut down.

Interpretation. This is a very useful simplification. Instead of solving an intractable many-product dynamic problem directly, the paper reduces the firm's problem to the values of individual product lines.

3.8 Optimal innovation intensity and shutdown thresholds

The optimal innovation intensity of a type- k firm is

$$x_k = \theta_k \left(\frac{(1 - \gamma) \mathbb{E}[\Upsilon_k(\hat{q} + \lambda \tilde{q})]}{\tilde{w}_s} \right)^{\frac{1-\gamma}{\gamma}}, \quad k \in \{l, h\}. \quad (11)$$

The shutdown threshold for a type- k firm is

$$\hat{q}_{k,\min} = \left(\frac{\tilde{w}_s \phi - \Omega_k}{\Pi} \right)^{\frac{1}{\varepsilon-1}}, \quad k \in \{l, h\}, \quad (12)$$

where Ω_k is the R&D value of an active product line and Π collects constant markup terms from the CES demand system.

Interpretation. High-type firms have larger franchise values and therefore lower shutdown thresholds. They keep lower-productivity product lines alive because each active line is a more valuable platform for future R&D.

3.9 Labor-market clearing

Unskilled labor is used in production and skilled labor is used in both operations and R&D. The skilled-labor clearing condition is

$$G(x_{entry}, \theta_E) + \sum_{k \in \{h, l\}} \Phi_k [h_k(w_s) + \phi] = L^S. \quad (13)$$

Interpretation. This is the core scarcity equation in the paper. Any policy that increases the mass of active incumbents also raises the amount of skilled labor tied up in fixed operating costs. That can crowd out R&D.

3.10 Aggregate growth

The growth rate of the economy is

$$g = \lambda\tau, \quad (14)$$

where τ is the average creative destruction rate.

Interpretation. Growth depends on two objects only:

- how often product lines are replaced,
- and how large the quality jump is when replacement occurs.

Policy works by changing who innovates, who exits, and therefore the creative-destruction rate.

3.11 Welfare

In stationary equilibrium, welfare can be written as

$$U_0(C_0, g) = \frac{1}{1 - \vartheta} \left[\frac{\Phi_0^{\frac{1-\vartheta}{\varepsilon-1}}}{\rho - (1 - \vartheta)g} - \frac{1}{\rho} \right], \quad (15)$$

with $C_0 = Q_0 = \Phi_0^{1/(\varepsilon-1)}$.

The paper compares policies using consumption-equivalent welfare changes.

Interpretation. Welfare depends not only on growth, but also on the initial mass of active product lines. So a policy can raise long-run growth while still having a limited welfare gain if it reduces current consumption too much. This is why the paper reports both growth and welfare.

3.12 Estimation by simulated method of moments

Five parameters are calibrated:

$$\varepsilon = 2.9, \quad L^S = 0.166, \quad \gamma = 0.5, \quad \vartheta = 2, \quad \rho = 0.02. \quad (16)$$

Eight parameters are estimated by SMM:

$$\{\phi, \theta_h, \theta_l, \theta_E, \alpha, \nu, \lambda, \varphi\}. \quad (17)$$

The objective function is

$$\min \sum_{i=1}^{18} \frac{|model(i) - data(i)|}{\frac{1}{2}|model(i)| + \frac{1}{2}|data(i)|}. \quad (18)$$

The model is solved numerically as a fixed point in a vector of six aggregate equilibrium objects, and firm-level moments are computed from a simulation of 2^{17} firms over 10,000 periods, where each period is 0.02 years.

Interpretation. The SMM step is where the paper turns from theory to quantitative policy analysis. The credibility of the later counterfactuals depends on whether this estimated model fits both the targeted and untargeted moments reasonably well.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the structural parameters

This is not an IV paper. Identification comes from the discipline imposed by the structural model together with the rich set of targeted moments.

A rough mapping from moments to parameters is:

- **Fixed cost of operation ϕ :** pinned down by exit patterns and the fixed-cost-to-R&D labor ratio.
- **Innovative capacities θ_h, θ_l :** pinned down by R&D intensities and growth differences across age-size groups.
- **Entrant innovative capacity θ_E :** pinned down by entrant behavior and the entrant employment share.
- **Probability of being high type α and decay rate ν :** pinned down by the life-cycle pattern that young firms are more innovative and high-growth than older firms.
- **Step size λ :** pinned down by the relationship between innovation, creative destruction, and aggregate growth.
- **Exogenous destruction φ :** pinned down by firm exit, especially in groups where endogenous obsolescence is less dominant.

Interpretation. The model is overidentified in a practical sense: 18 moments discipline 8 estimated parameters. So the later policy results are not coming from a knife-edge calibration to one or two moments.

4.2 Why the selection mechanism is empirically central

The model is built to explain two facts jointly:

- some firms do much more R&D and grow faster than others,
- but low-productivity or low-type product lines also exit endogenously.

The key mechanism is the interaction of:

- heterogeneous innovative capacity,

- endogenous exit thresholds,
- and type transitions from high to low.

Interpretation. If the model fit came only from average R&D or average growth, the policy lessons would be weak. What matters is that it also matches the *life-cycle composition* of firms and the selection margin.

4.3 Validation through nontargeted moments

The paper checks the estimated model against moments not used in estimation:

- persistence in employment growth and R&D intensity,
- R&D per employee and patents per employee for high versus low management-score firms,
- productivity dispersion within size-age bins,
- product-line distributions,
- and a growth decomposition into within, between, cross, entry, and exit components.

Interpretation. This is the paper's equivalent of external validity inside the sample. If the model had matched the 18 target moments but failed badly on these auxiliary facts, the policy counterfactuals would be much less persuasive.

4.4 Why reverse causality is not the main issue here

The paper is not estimating a reduced-form causal effect of one observed variable on another. Instead, it estimates primitives of a stationary equilibrium model and then computes counterfactual stationary equilibria.

Interpretation. The main risk is not classic reverse causality. The risk is *model misspecification*: if the two-type structure, the cost functions, or the labor-market bottleneck are badly chosen, the policy conclusions could be wrong even if the model fits the moments mechanically.

4.5 What the paper can and cannot distinguish

The paper can distinguish relatively well between policies that:

- encourage innovation,
- preserve incumbents,
- or free up skilled labor from incumbent operations.

But it cannot fully distinguish:

- all possible sources of firm heterogeneity beyond the high/low type structure,
- transitional dynamics of policy changes, since welfare is evaluated across stationary equilibria,
- or the decision of continuing noninnovative firms to become innovative from zero.

It also keeps monopoly pricing distortions in place when studying planner allocations, because the focus is on dynamic R&D and reallocation distortions rather than on static markup distortions.

Interpretation. The paper is best read as a strong quantitative argument about *innovation misallocation through firm selection*, not as a complete quantitative welfare analysis of every distortion in manufacturing.

4.6 Why the policy message is nevertheless credible

The most robust pattern in the policy exercises is not a fine numerical point estimate. It is a ranking:

- subsidizing incumbent operations is bad,
- subsidizing entry alone is weak,
- subsidizing incumbent R&D helps only modestly,
- pushing low-type firms toward exit is powerful.

This ranking survives employment weighting, exclusion of M&A-related activity, inclusion of non-innovative firms, alternative fixed-cost technologies, reallocation costs, more than two firm types, and endogenous skill supply.

Interpretation. Even if one does not trust every decimal place, the qualitative lesson is hard to miss: the paper is really about the power of selection, not just the level of R&D.

5 Tables and figures

5.1 Table 1 and Table 2: calibrated and estimated parameters

Parameter	Description	Value
1. ε	Constant elasticity of substitution	2.900
2. L^S	Measure of high-skilled workers	0.166
3. γ	Innovation elasticity	0.500
4. ϑ	Inverse of the intertemporal elasticity of substitution	2.000
5. ρ	Discount rate	0.020

Parameter	Description	Value	Standard error
1. ϕ	Fixed cost of operation	0.216	(0.012)
2. θ^h	Innovative capacity of high-type firms	1.751	(0.020)
3. θ^l	Innovative capacity of low-type firms	1.391	(0.017)
4. θ^E	Innovative capacity of entrants	0.024	(0.001)
5. α	Probability of being high-type entrant	0.926	(0.023)
6. ν	Transition rate from high-type to low-type	0.206	(0.005)
7. λ	Innovation step size	0.132	(0.010)
8. φ	Exogenous destruction rate	0.037	(0.001)

Read:

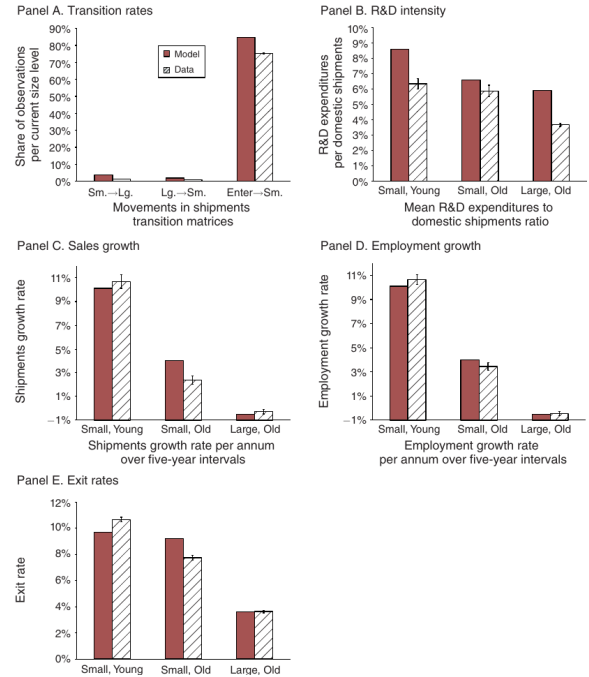
- Table 1 fixes the calibrated parameters at economically standard values:
 - $\varepsilon = 2.9$,
 - $L^S = 0.166$,
 - $\gamma = 0.5$,
 - $\vartheta = 2$,
 - $\rho = 0.02$.

- Table 2 estimates the remaining primitives:
 - fixed cost $\phi = 0.216$,
 - $\theta_h = 1.751$ and $\theta_l = 1.391$,
 - entrant innovative capacity $\theta_E = 0.024$,
 - high-type entrant probability $\alpha = 0.926$,
 - transition rate $\nu = 0.206$,
 - innovation step size $\lambda = 0.132$,
 - exogenous destruction $\varphi = 0.037$.

Economic message. The parameter estimates already tell the story. Entrants are very likely to start as high-type firms, but high types decay into low types fairly quickly. That creates a strong motive for policy to discriminate across incumbents by how innovative they still are—but uniform policies cannot observe type directly.

5.2 Table 3 and Figure 1: targeted moments and goodness of fit

	Moments	Model	Data	Standard error
1.	Firm exit (small-young)	0.097	0.107	(0.002)
2.	Firm exit (small-old)	0.092	0.077	(0.002)
3.	Firm exit (large-old)	0.036	0.036	(0.001)
4.	Transition from large to small	0.021	0.010	(0.001)
5.	Transition from small to large	0.038	0.014	(0.001)
6.	Probability of small (conditional on entry)	0.848	0.753	(0.005)
7.	Employment growth (small-young)	0.101	0.106	(0.004)
8.	Employment growth (small-old)	0.040	0.035	(0.003)
9.	Employment growth (large-old)	−0.005	−0.005	(0.002)
10.	Sales growth (small-young)	0.101	0.107	(0.006)
11.	Sales growth (small-old)	0.040	0.024	(0.004)
12.	Sales growth (large-old)	−0.005	−0.003	(0.002)
13.	R&D to sales (small-young)	0.086	0.064	(0.004)
14.	R&D to sales (small-old)	0.066	0.059	(0.001)
15.	R&D to sales (large-old)	0.059	0.037	(0.001)
16.	Five-year entrant share	0.336	0.393	(0.003)
17.	Fixed cost–R&D labor ratio	4.175	5.035	(0.015)
18.	Aggregate growth	0.023	0.022	(0.007)



Read:

- Table 3 reports the 18 targeted moments and shows that the model matches the data closely across exit, transition, growth, R&D intensity, entrant shares, and aggregate growth.
- Figure 1 visualizes the fit by comparing model versus data for:
 - transition rates,
 - R&D intensity,

- sales growth,
 - employment growth,
 - exit rates.
- The fit is especially good for large-old exit rates, young small-firm growth, and aggregate growth.

Economic message. This is the paper’s first major success. The policy analysis is only worth taking seriously if the model can replicate the joint age-size dynamics of innovative firms, and Figure 1 shows that it broadly does.

5.3 Table 4 and Figure 2: the baseline economy and the selection margin

Read:

- Table 4 summarizes the baseline equilibrium:
 - entrant innovation flow $x^{entry} = 0.51$,
 - low-type innovation intensity $x^l = 25.90$,
 - high-type innovation intensity $x^h = 38.13$,
 - share of product lines operated by low types $\Phi^l = 55.04$,
 - share operated by high types $\Phi^h = 6.28$,
 - low-type shutdown threshold $\hat{q}_{l,min} = 147.26$,
 - high-type shutdown threshold $\hat{q}_{h,min} = 130.33$,
 - skilled labor in R&D as a share of skilled labor supply: 19.86,
 - creative destruction rate $\tau = 17.16$,
 - growth $g = 2.26$,
 - welfare normalized to 100.
- Figure 2 shows why selection matters: the shutdown threshold is lower for high-type firms than for low-type firms.

Economic message. High-type firms keep product lines alive at lower relative productivity because each active line is a more valuable springboard for future R&D. That is the central reason why pushing low-type firms out of operation can raise both innovation and welfare.

5.4 Table 5: nontargeted moments

Read:

- Panel A checks persistence: the model produces higher future employment growth and higher R&D intensity for the top 10% of firms than for the bottom 90%.
- Panel B compares high-versus-low management firms:
 - R&D per employee ratio: 1.737 in the model versus 1.461 in the data,

TABLE 4—BASELINE ECONOMY										
x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
0.51	25.90	38.13	55.04	6.28	147.26	130.33	19.86	17.16	2.26	100

Both Table 3 and Figure 1 show a relatively good fit between our model implied

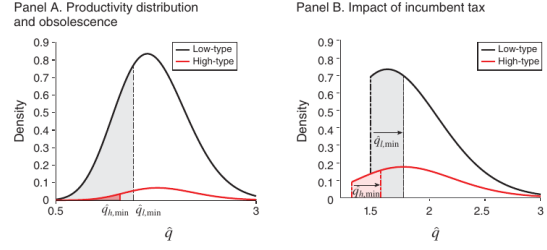


FIGURE 2. PRODUCTIVITY DISTRIBUTION AND SELECTION

TABLE 5—NONTARGETED MOMENTS			
Moments	Model	Data mean	Standard error
<i>Panel A</i>			
Employment growth of bottom 90 percent	0.011	0.011	(0.004)
Employment growth of top 10 percent	0.016	0.037	(0.016)
R&D to sales ratio of bottom 90 percent	0.061	0.038	(0.004)
R&D to sales ratio of top 10 percent	0.071	0.052	(0.018)
<i>Panel B</i>			
R&D per employee ratio (high to low)	1.737	1.461	(n/a)
Patent per employee (high to low)	1.578	1.838	(n/a)
<i>Panel C</i>			
Productivity distribution, 75/25: small/young	1.328	1.344	(0.006)
Productivity distribution, 75/25: small/old	1.231	1.311	(0.006)
Productivity distribution, 75/25: large/young	1.150	1.388	(0.017)
Productivity distribution, 75/25: large/old	1.087	1.294	(0.004)
<i>Panel D</i>			
Add a product (or more)	2.9%	8.0%	(0.003)
Drop a product (or more)	2.3%	8.3%	(0.003)
Do both	91.7%	76.8%	(0.005)

- patents per employee ratio: 1.578 in the model versus 1.838 in the data.
- Panel C compares 75/25 productivity dispersion across size-age bins and gets similar patterns.
- Panel D compares product-line adjustment events and finds broadly reasonable magnitudes.

Economic message. These are important out-of-sample checks. The model is not merely reproducing the exact moments it was fed; it is also generating sensible life-cycle and productivity patterns that resemble the data.

5.5 Figure 3 and Table 6: product-line distribution and growth decomposition

Read:

- Figure 3 compares the model-implied product-line distribution to the empirical five-digit and seven-digit product distributions and finds a close match.
- Table 6 decomposes growth:
 - within share: 0.607 in the model versus 0.999 in the data,
 - between share: -0.024 versus -0.049 ,
 - cross share: 0.239 versus -0.305 ,
 - entry share: 0.175 versus 0.192,
 - exit share: 0.003 versus 0.164,
 - ten-year cumulative growth: 0.254 versus 0.261.

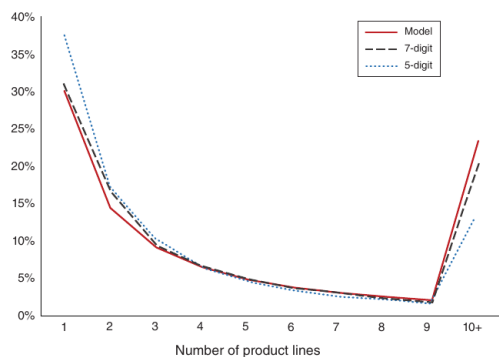


FIGURE 3. PRODUCT LINE DISTRIBUTION

TABLE 6—DECOMPOSITION

	Model	Data	Standard error
Within share	0.607	0.999	(0.176)
Between share	−0.024	−0.049	(0.057)
Cross share	0.239	−0.305	(0.176)
Entry share	0.175	0.192	(0.062)
Exit share	0.003	0.164	(0.033)
Net entry share	0.178	0.356	(0.074)
Ten-year cumulative growth	0.254	0.261	(0.051)

up over the components too. The one exception is in the cross term, where the model finds growing share of employment connected to growing labor productivity whereas the data finds a negative correlation. This discrepancy is not a very robust feature, however; the cross term becomes positive, for example, when we look at the

Economic message. The fit is not perfect—especially for the cross term—but the model gets the broad decomposition right: within-firm productivity growth is the dominant component, and entry also matters materially. This is enough to make the policy exercises informative about the role of reallocation.

5.6 Tables 7–9: simple industrial policies one at a time

TABLE 8—OPERATION SUBSIDY

x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
<i>Panel A. Baseline</i>										
0.51	25.90	38.13	55.04	6.28	147.26	130.33	19.86	17.16	2.26	100.00
<i>Panel B. 1 percent of GDP ($s_o = 4\%$)</i>										
0.50	25.57	37.58	55.63	6.05	146.27	129.30	19.39	17.00	2.24	99.80

TABLE 9—ENTRY SUBSIDY

x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
<i>Panel A. Baseline</i>										
0.51	25.90	38.13	55.04	6.28	147.26	130.33	19.86	17.16	2.26	100.00
<i>Panel B. 1 percent of GDP ($s_e = 65\%$)</i>										
1.35	24.59	35.08	49.50	10.16	151.30	138.46	22.02	17.09	2.25	98.36

TABLE 10—SOCIAL PLANNER

x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
<i>Panel A. Baseline</i>										
0.51	25.90	38.13	55.04	6.28	147.26	130.33	19.86	17.16	2.26	100.00
<i>Panel B. Social planner</i>										
0.60	25.42	45.34	5.64	44.70	240.42	27.80	34.21	22.30	2.94	104.47

The social planner improves growth and welfare quite significantly. Growth increases from 2.26 percent to 2.94 percent. Welfare increases by 4.47 percent, underscoring that the equilibrium was far from optimal in the baseline model, and

Read:

- **Table 7: incumbent R&D subsidy.** A policy worth 1% of GDP corresponds to a 14% subsidy to incumbent R&D. It raises incumbent innovation, but also raises skilled wages and crowds out entry. Growth rises only from 2.26% to 2.34%, and welfare rises only to 100.63.

- **Table 8: operation subsidy.** A 1% of GDP subsidy to fixed operating costs corresponds to a 4% subsidy. It discourages exit, raises the share of product lines run by low-type firms, lowers growth to 2.24%, and lowers welfare to 99.80.
- **Table 9: entry subsidy.** A 1% of GDP subsidy to entrants corresponds to a 65% entry subsidy. It raises entrant innovation flow strongly, but reduces incumbent innovation and slightly lowers growth to 2.25% and welfare to 98.36.

Economic message. This is the paper’s cleanest policy ranking. More subsidies do not automatically mean more growth. If a subsidy interferes with selection or reallocation, it can be close to useless or even harmful.

5.7 Tables 10–12: the social planner and optimal uniform policies

	x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	g	Wel
1. Baseline	0.51	25.90	38.13	55.04	6.28	147.26	130.33	2.26	100.00
2. Social planner (SP)	0.60	25.42	45.34	5.64	44.70	240.42	27.80	2.94	104.47
3. SP choosing innovation	0.52	25.63	38.71	54.45	6.91	147.26	130.33	2.26	100.00
4. SP choosing $\hat{q}_{min}$	0.94	25.90	38.13	39.74	18.92	161.16	29.91	2.43	101.58

	x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
<i>Panel A. Baseline</i>	0.51	25.90	38.13	55.04	6.28	147.26	130.33	19.86	17.16	2.26	100.00
<i>Panel B. Incumbent R&D ($s_i = 39\%$)</i>	0.38	30.74	46.54	47.67	8.65	160.07	142.83	26.40	19.06	2.51	101.22
<i>Panel C. Operation ($s_o = -69\%$)</i>	0.61	30.78	46.04	45.95	9.84	161.50	145.72	27.08	19.29	2.54	101.42
<i>Panel D. Entry ($s_e = 18\%$)</i>	0.62	25.74	37.69	54.26	6.95	147.58	131.35	20.00	17.20	2.27	100.04
<i>Panel E. Incumbent R&D and Operation ($s_i = -3\%$, $s_o = -74\%$)</i>	0.63	30.74	45.94	45.90	9.90	161.50	145.81	27.07	19.29	2.54	101.42

Figure 1: Tables 11 and 12 from Acemoglu et al. (2018).

Read:

- **Table 10: social planner.** The planner raises growth from 2.26% to 2.94% and welfare from 100 to 104.47. The low-type exit threshold jumps sharply upward and the high-type threshold drops sharply, while high-type R&D rises strongly.
- **Table 11: restricted planner.** If the planner can only change innovation and entry while keeping exit thresholds fixed, welfare barely moves. If she can only change exit thresholds while keeping innovation fixed, welfare still rises to 101.58 and growth to 2.43.
- **Table 12: optimal uniform policies.**
 - incumbent R&D subsidy alone: $s_i = 39\%$ gives $g = 2.51$ and welfare = 101.22;
 - operation tax alone: $s_o = -69\%$ gives $g = 2.54$ and welfare = 101.42;
 - entry subsidy alone: almost no gain;
 - optimal combination: a tiny tax on incumbent R&D ($s_i = -3\%$) and a large tax on operation costs ($s_o = -74\%$) also yields $g = 2.54$ and welfare = 101.42.

Economic message. The surprising result is that the planner’s power comes mostly through the *exit margin*, not through simply pushing up innovation effort. The useful policy is therefore one that frees skilled labor from weak incumbents, not one that gives a blanket subsidy to all R&D.

5.8 Tables 13–19: robustness

<i>Panel A. Baseline</i>									
0.52	25.26	47.76	63.37	11.62	126.53	89.02	23.86	2.22	100.00
<i>Panel B. Social planner</i>									
0.58	24.43	52.99	38.08	28.90	152.47	42.06	31.99	25.20	101.25
<i>Panel C. Incumbent R&D and operation ($s_i = 0\%$, $s_o = -47\%$)</i>									
0.57	27.28	51.53	53.96	16.41	138.78	109.77	28.54	23.74	100.56

B. Organic Sample that Excludes M&A Activities

Our baseline sample includes “inorganic” entry and exit, taking the form of mergers and acquisitions (M&A) and spinouts (where part of an existing firm becomes a new legal entity). We next reestimate the model after removing all observations we determined to be potentially influenced by inorganic activity on these margins.¹⁸ The results from this exercise are reported in Table 14. The broad patterns of various policy implications remain very similar to the baseline: for example, the social planner is now able to increase growth from 2.24 percent to 2.90 percent, with a 4.17 percent consumption-equivalent welfare gain, and the optimal policies once

Panel A. Baseline										
0.46	27.24	33.97	55.48	2.41	154.57	146.64	18.33	16.39	2.24	100.00
Panel B. Social planner										
0.58	28.84	44.41	3.17	44.38	269.34	29.39	32.91	21.20	2.90	104.17
Panel C. Incumbent R&D and operation ($s_i = -4\%$, $s_o = -84\%$)										
0.60	33.99	43.00	48.80	3.49	168.45	160.26	26.22	18.69	2.56	101.82

TABLE 15—FULL MANUFACTURING (Non-Innovating Firms Included)

x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
<i>Panel A. Baseline</i>										
1.71	5.08	29.69	22.92	4.63	215.50	118.64	25.47	4.25	1.92	100.00
<i>Panel B. Social planner</i>										
1.95	5.29	35.03	16.93	6.63	256.76	53.92	36.29	5.17	2.34	101.88
<i>Panel C. Incumbent R&D and operation ($s_i = 7\%$, $s_o = -41\%$)</i>										
1.80	5.72	33.80	20.34	5.10	233.44	149.89	31.17	4.69	2.12	100.75

Read:

- **Employment-weighted moments (Table 13).** Baseline growth is 2.22%. The planner raises it to 2.54% with welfare 101.25. Uniform policies still help, though by less.
- **Excluding M&A-related activity (Table 14).** Baseline growth is 2.24%, planner growth is 2.90%, and welfare rises to 104.17. The message is unchanged.
- **Full manufacturing sample including noninnovative firms (Table 15).** Baseline growth falls to 1.92%, but the planner still raises it to 2.34% and uniform policies still help modestly.
- **Unskilled overhead labor (Table 16).** Results remain qualitatively similar, but gains shrink because operation taxes free less skilled labor.
- **Reallocation cost (Table 17).** With training/reallocation costs, baseline growth is 2.25%, planner growth is 2.77%, and optimal uniform policies still generate a welfare gain of 1.44%.
- **Three firm types (Table 18).** The planner raises growth from 2.20% to 2.94% and welfare to 105.60; the main selection logic survives.
- **Endogenous skill supply (Table 19).** Education subsidies become important, but even after optimally subsidizing education, the same selection-based operation tax logic continues to raise growth and welfare.

Economic message. The quantitative gains move around across robustness exercises, but the *ranking of policies* remains strikingly stable. That gives the paper much more credibility than a result that only worked in one exact benchmark calibration.

6 Short summary

- The paper builds a firm-dynamics model with endogenous innovation, creative destruction, entry, exit, and heterogeneity in innovative capacity.

TABLE 17—MODEL WITH REALLOCATION COST

x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
<i>Panel A. Baseline</i>										
0.41	23.49	43.60	60.26	4.89	137.43	106.61	19.57	16.70	2.25	100.00
<i>Panel B. Social planner</i>										
0.49	24.53	52.38	31.57	23.58	173.87	30.47	31.65	20.58	2.77	102.55
<i>Panel C. Incumbent R&D and operation ($s_l = 7\%$, $s_h = -73\%$)</i>										
0.46	26.55	52.72	48.15	10.81	154.34	127.68	27.05	18.94	2.55	101.44

TABLE 18—MODEL WITH THREE TYPES ($\theta^H, \theta^M, \theta^L$)

x^{entry}	x^l	x^m	x^h	Φ^l	Φ^m	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{m,min}$	$\hat{q}_{h,min}$	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
<i>Panel A. Baseline</i>													
0.51	25.83	38.19	40.85	52.32	5.07	1.12	153.14	136.78	132.04	19.11	16.41	2.20	100.00
<i>Panel B. Social planner</i>													
0.34	23.23	45.83	46.97	2.30	24.21	21.19	277.38	0.25	95.08	34.04	21.92	2.94	105.60
<i>Panel C. Incumbent R&D and operation ($s_l = 6\%$, $s_m = -69\%$)</i>													
0.59	31.35	46.90	50.83	42.30	6.50	3.78	169.22	154.47	149.30	27.29	18.82	2.52	101.81

TABLE 19—MODEL WITH ENDOGENOUS SUPPLY OF SKILLS

x^{entry}	x^l	x^h	Φ^l	Φ^h	$\hat{q}_{l,min}$	$\hat{q}_{h,min}$	L^S	$\frac{L^{R\&D}}{L^S}$	τ	g	Wel
<i>Panel A. Baseline</i>											
0.51	25.90	38.13	55.04	6.28	147.26	130.33	16.55	19.86	17.16	2.26	100.00
<i>Panel B. Baseline with optimal education subsidy ($s_{edu} = 81\%$)</i>											
0.55	27.46	40.98	60.02	8.26	133.97	114.56	18.94	22.03	20.41	2.69	110.96
<i>Panel C. Social planner</i>											
0.63	26.81	47.73	16.13	41.68	188.74	28.86	18.93	33.93	24.84	3.28	114.42
<i>Panel D. Incumbent R&D, operation cost, and education policies ($s_l = -3\%$, $s_m = -62\%$, $s_{edu} = 92\%$)</i>											
0.65	31.32	47.22	50.58	12.36	147.01	129.78	18.94	28.10	22.33	2.94	112.08

- High-type firms are better innovators than low-type firms, but high types can transition into low types over time.
- Operating a product line requires skilled labor, so the economy faces a misallocation problem: too much skilled labor is tied up in operations and too little is used in R&D.
- The model is estimated with SMM on detailed U.S. Census data and matches targeted and untargeted moments well.
- Blanket subsidies to incumbent R&D raise growth only modestly.
- Subsidies to incumbents' operating costs are harmful because they prop up low-type firms and worsen selection.
- A social planner can achieve large gains mainly by changing exit thresholds and reallocating skilled labor toward high-type firms.
- The best realistic uniform policy is a large tax on the operation of incumbents, not a blanket R&D subsidy.
- The policy message is robust across many variations of the sample and model.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the main inefficiency in an innovation economy with heterogeneous firms is not just underinvestment in R&D, but *misallocation of skilled labor across firms*. Because low-type incumbents absorb skilled labor in operations while contributing relatively little to future innovation, policies that keep them alive can depress growth. In contrast, policies that force low-type product lines toward exit and reallocate skilled labor toward high-type firms produce substantial gains.

7.2 Economic intuition

The logic is simple but powerful.

- Every active product line uses skilled labor just to stay in operation.
- But only some firms are really good at turning skilled labor into future innovations.
- A uniform R&D subsidy helps all incumbents, including weak ones.
- An operation tax, by contrast, bites hardest on low-type firms close to the exit margin.

So the operation tax acts like a coarse but effective selection device. It pushes the economy toward a composition where scarce skilled labor is attached to better innovators.

7.3 Takeaway

This paper is a strong reminder that growth policy should be designed around *who innovates* and *who exits*, not only around how much money is spent on R&D. In this model, the best way to get more innovation is not to subsidize every incumbent. It is to let weak incumbents contract or exit and to free up resources for the firms that can use them best.